

Comment

NEET 2024

FAST TRACK SERIES

BIOMOLECULES



PART 2



Q2: Malonate inhibits the growth of pathogenic bacteria by inhibiting the activity of [2023]

1. Dinitrogenase
2. Succinic dehydrogenase
3. Amylase
4. Lipase

Q3: Given below are two statements:

Statement I: Low temperature preserves the enzyme in a temporarily inactive state whereas high temperature destroys enzymatic activity because proteins are denatured by heat.

Statement II: When the inhibitor closely resembles the substrate in its molecular structure and inhibits the activity of the enzyme, it is known as competitive inhibitor

In the light of the above statements, choose the correct answer from the options given below:

[2023]

1. Statement I is false but Statement II is true.
2. Both Statement I and Statement II are true.
3. Both Statement I and Statement II are false.
4. Statement I is true but Statement II is false.

Q4: Given below are two statements:

Statement I: A protein is imagined as a line, the left end represented by first amino acid (C-terminal) and the right end represented by last amino acid (N-terminal)

Statement II: Adult human haemoglobin, consists of 4 subunits (two subunits of α type and two subunits of β type.)

In the light of the above statements, choose the correct answer from the options given below:

[2023]

1. Statement I is false but Statement II is true.
2. Both Statement I and Statement II are true.
3. Both Statement I and Statement II are false.
4. Statement I is true but Statement II is false.

PYQS

Q6: Which of the following is not a secondary metabolite?

[2023]

1. Curcumin
2. Morphine
3. Anthocyanin
- ✓ 4. Lecithin

PYQS

Q7: Match List - I with List - II

List - I

- (A) Terpenoides
- (B) Unsaturated fatty acid
- (C) Nucleic acid
- (D) Polysaccharide

List - II

- (I) Codeine
- (II) Diterpenes
- (III) Ricin
- (IV) Concanavalin

Choose the correct answer from the options given below

[2023]

- | | (A) | (B) | (C) | (D) |
|----|-----|-----|-----|-----|
| 1. | II | IV | III | I |
| 2. | II | I | IV | III |
| 3. | II | III | I | IV |
| 4. | II | IV | I | III |

PYQS

Q8: Exoskeleton of arthropods is composed of:

[2022]

1. Glucosamine
2. Cutin
3. Cellulose
- ✓ 4. Chitin

PYQS

Q9: Read the following statements on lipids and find out correct set of statements:

- a. Lecithin found in the plasma membrane is a glycolipid.**
- b. Saturated fatty acids possess one or more $C=C$ bonds.**
- c. Gingelly oil has lower melting point, hence remains as oil in winter.**
- d. Lipids are generally insoluble in water but soluble in some organic solvents.**
- e. When fatty acid is esterified with glycerol, monoglycerides are formed**

[2022]

- 1. (a), (b) and (d) only
- 2. (a), (b) and (c) only
- 3. (a), (d) and (e) only
- 4. (c), (d) and (e) only

PYQS

Q10: A dehydration reaction links two glucose molecules to produce maltose. If the formula for glucose is $C_6H_{12}O_6$ then what is the formula for maltose?

[2022]

1. $C_{12}H_{24}O_{11}$
2. $C_{12}H_{20}O_{10}$
3. $C_{12}H_{24}O_{12}$
4. $C_{12}H_{22}O_{11}$

PYQS

Q11: Match List - I with List - II

List - I

- (a) Glycogen
- (b) Globulin
- (c) Steroids
- (d) Thrombin

List - II

- (i) Hormone
- (ii) Biocatalyst
- (iii) Antibody
- (iv) Storage product

Choose the correct answer from the options given below

[2022]

- | | (a) | (b) | (c) | (d) |
|----|-----|-----|-----|-----|
| 1. | iv | iii | i | ii |
| 2. | iii | ii | iv | i |
| 3. | iv | ii | i | iii |
| 4. | ii | iv | iii | i |

PYQS

Q:12 Given below are two statements:

Statement I: Cellulose is a polymeric polysaccharide

T

Statement II: The building blocks of cellulose are glucose molecules

T

In the light of the above statements, choose the correct answer from the options given below:

1. Statements-I is correct but statements-II is incorrect.
2. Statements-I is incorrect but statements-II is correct.
3. Both statements-I and statements-II are correct.
4. Both statements-I and Statements -II are false.

β -glucose

[2022]

PYQS

Q13: Which of the following bond is formed as a result of reaction of carboxyl group of one amino acid with amino group of other amino acids with elimination of water?

[2022]

1. Phosphodiester Bond
2. Hydrogen Bond
3. Glycosidic Bond
4. Peptide Bond

Q14: In a reaction catalyzed by an enzyme, which of the following statements is correct?
[2022]

1. Enzymes decrease the activation energy for formation of transition state.
2. Enzymes make transition from substrate to product more difficult.
3. Enzymes increase the activation energy for formation of transition state.
4. Enzyme-substrate complex formed during a reaction lasts for a very long time.

PYQS

Q15: Match List - I with List - II

List - I

- (A) Adenine
- (B) Anthocyanin
- (C) Chitin
- (D) Codeine

List - II

- (I) Pigment
- (II) Polysaccharide
- (III) Alkaloid
- (IV) Purine

Choose the correct answer from the options given below

[2022]

- | | (A) | (B) | (C) | (D) |
|----|-----|-----|-----|-----|
| 1. | IV | I | II | III |
| 2. | IV | III | II | I |
| 3. | III | I | IV | II |
| 4. | I | IV | III | II |

Q16: Primary proteins are also known as polypeptides, because:

[2022]

1. they are linear chains.
2. they are polymers of peptide monomers.
3. successive amino acids are joined by peptide bonds.
4. they can assume many conformations.

Q:17 Given below are two statements:

Statement I: Amino acids have a property of ionizable nature of -NH_2 and -COOH groups, hence have different structures at different pH

Statement II: Amino acids can exist as Zwitterionic form at acidic and basic pH.

In the light of the above statements, choose the correct answer from the options given below:

[2022]

1. Both Statement I and Statement II are correct.
2. Both Statement I and Statement II are Incorrect.
3. Statement I is correct but Statement II is incorrect.
4. Statement I is incorrect but Statement II is correct.

PYQS

Q18: If A and C make 30% and 20% of DNA, respectively, what will be the percentage composition of T and G?

$$\begin{aligned} A = 30 &= T = 30 \\ C = 20 &= G = 20 \end{aligned}$$

[2022]

1. T: 20%, G: 30%

✓ 2. T: 30%, G: 20%

3. T: 30%, G: 30%

4. T: 20%, G: 20%

PYQS

Q19: In the enzyme which catalyses the breakdown of:



The prosthetic group is:

[2022]

1. Nicotinamide adenine dinucleotide
2. Haem
3. Zinc
4. Niacin

PYQS

Q20: Which of the following are not secondary metabolites in plants?

[2021]

1. Vinblastin, curcumin
2. Rubber, gums
3. Morphine, codeine
4. Amino acids, glucose





PYQS

Q21: Identify the incorrect pair.

[2021]

1. Lectins - Concanavalin A
2. Drugs - Ricin
3. Alkaloids - Codeine
4. Toxin - Abrin

PYQS

Q22: Following are the statements with reference to 'lipids'.

(a) Lipids having only single bonds are called unsaturated fatty acids.

(b) Lecithin is a phospholipid.

(c) Trihydroxy propane is glycerol.

(d) Palmitic acid has 20 carbon atoms including carboxyl carbon.

(e) Arachidonic acid has 16 carbon atoms.

You have to choose the correct statements out of the five given above. Choose the correct answer from the options given below:

[2021]

1. (b) and (c) only

2. (b) and (e) only

3. (a) and (b) only

4. (c) and (d) only



PYQS

Q23: Which one is the most abundant protein in the animal world?

[2020]

1. Trypsin
2. Haemoglobin
3. Collagen
4. Insulin

Q24: Identify the substances having a glycosidic bond and peptide bond, respectively in their structure:

[2020]

1. Glycerol, trypsin
2. Cellulose, lecithin
3. Inulin, insulin
4. Chitin, cholesterol

PYQS

Q25: Identify the basic amino acid from the following. [2020]

1. Glutamic Acid
2. Lysine
3. Valine
4. Tyrosine